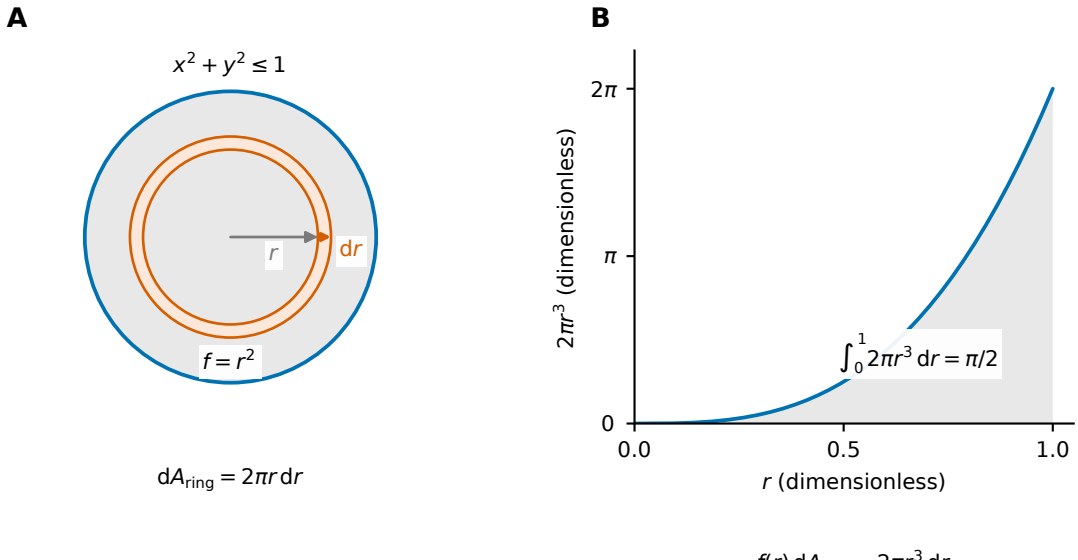
A
$x^2 + y^2 \leq 1$
r
$\mathrm{d}r$
$f = r^2$
$\mathrm{d}A_{\mathrm{ring}} = 2\pi r\,\mathrm{d}r$
B
2π
π
0
$2\pi r^3$ (dimensionless)
$\int_0^1 2\pi r^3\,\mathrm{d}r = \pi/2$
0.0
0.5
1.0
r (dimensionless)
$f(r)\,\mathrm{d}A_{\mathrm{ring}} = 2\pi r^3\,\mathrm{d}r$